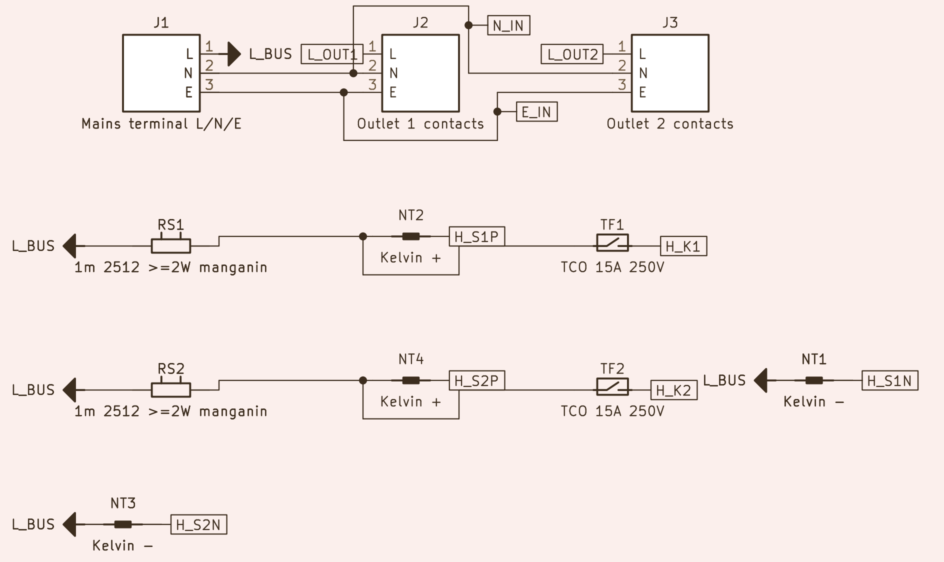
HOT SIDE

Mains referenced. Hot–side ground = L_BUS = LIVE.

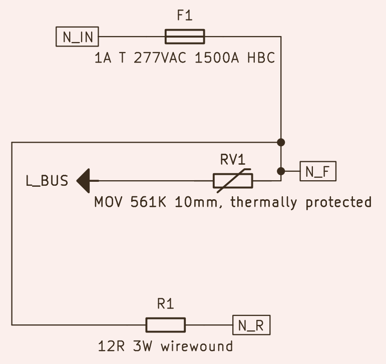
1. MAINS ENTRY AND OUTLET FEEDS

Ring L/N/E land on the busbar; only outlet current passes the shunts and relays.
Shunt common end = L_BUS = hot–side ground. Net ties NT1–NT4 force the IP/IN taps to start at the shunt pads (Kelvin).
Thermal cut–offs: non–resettable, clamped to the outlet feed near the contacts; Tf chosen after the EVT thermal test.



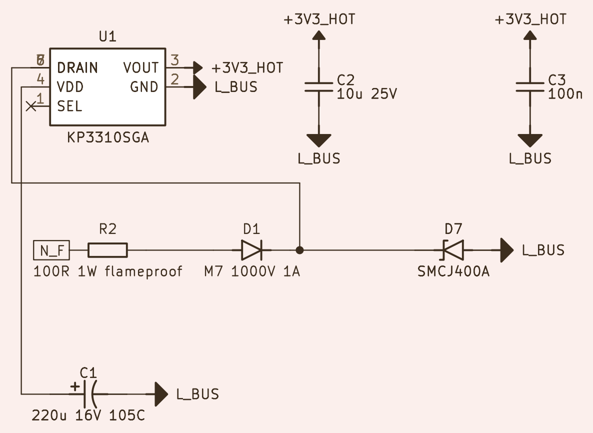
2. ELECTRONICS FEED AND SURGE PROTECTION

Fuse and series resistor per Mornsun's LD03 application circuit. MOV after the fuse, thermally protected type.
F1 is a 1500 A breaking–capacity fuse: if RV1 fails short, F1 must interrupt the circuit's prospective fault current.



3. HOT–SIDE 3.3 V SUPPLY (KP3310SGA)

Half–wave from N_F; KP3310 charges VDD (7.8 V) while the line is low, then an LDO makes 3.3 V. SEL open = 3.3 V.
D7 clamps the drain below its 650 V rating in a surge; R2 limits the clamp current (value from the R1 simulation).
Hold–up: 15 mA (worst case) x 20 ms / 220 uF = 1.4 V droop on VDD (7.8 V to 6.4 V), so VOUT stays in regulation.

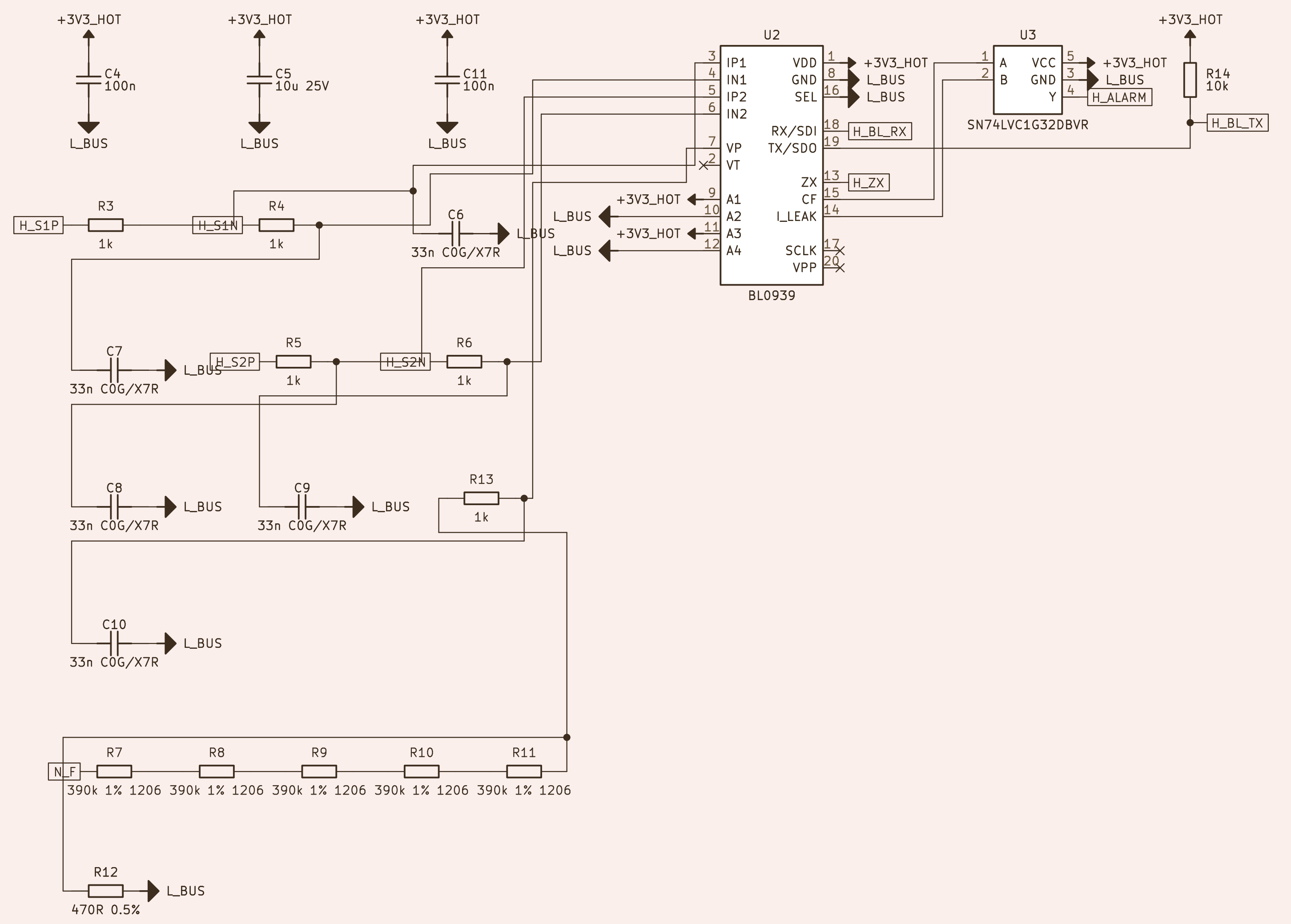


HOT SIDE, SENSING

Metering and wiring checks, all mains referenced.

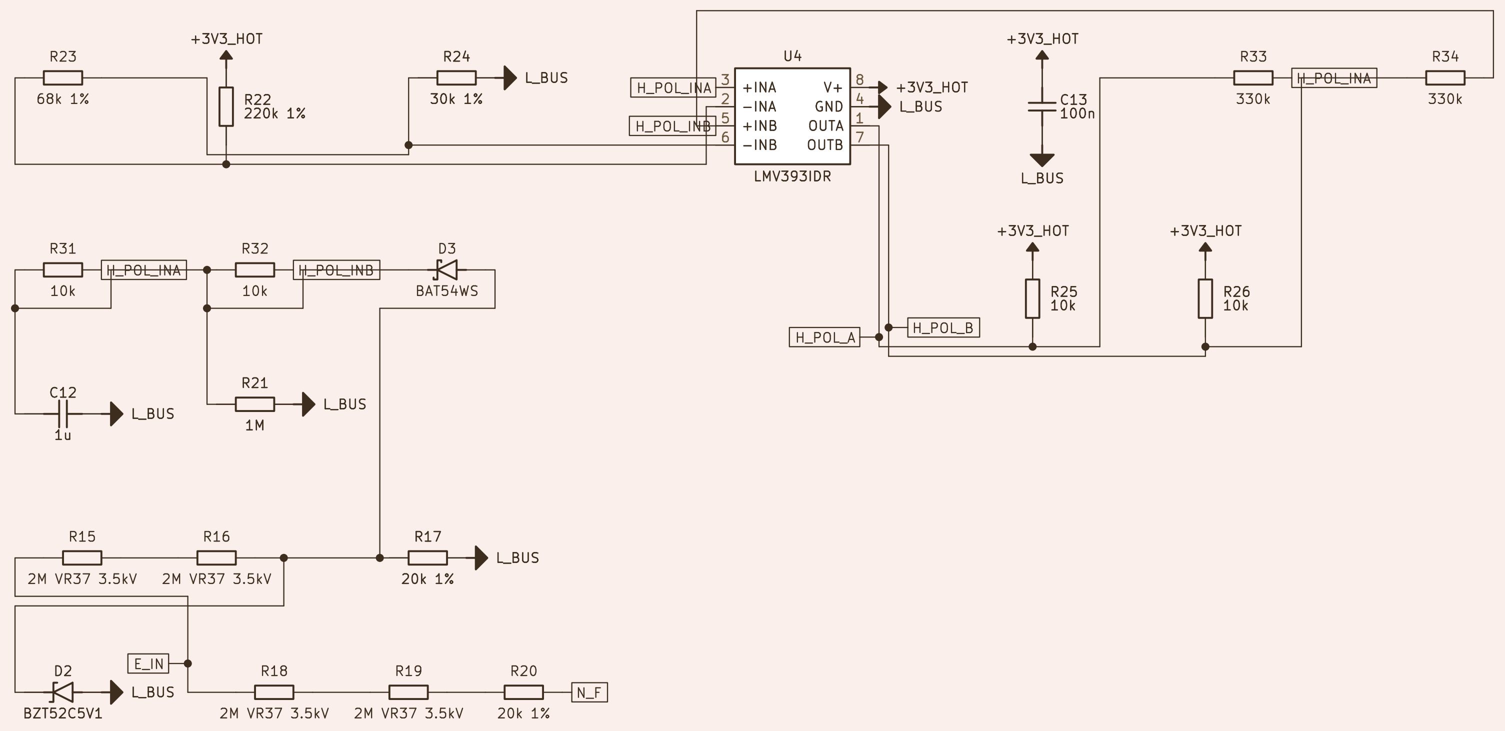
4. METERING (BL0939)

UART address 5 (A1, A3 high; A2, A4 low): the firmware sends 0x55/0xA5, which encode address 5.
Divider 5 x 390k + 470R: 230 V gives 55 mV rms, 290 V gives 70 mV rms (pin limit 70 mV rms). Firmware board file: div_r_bottom = 0.47 k.
Alarms: CF (channel A, MODE[12]) OR I_LEAK (channel B) –> isolator channel E. Both active high.



5. POLARITY AND EARTH CHECK

Arm A: E –> 2 x 2M (HV) –> sense 20k –> L_BUS. Arm B: E –> 2 x 2M (HV) + 20k –> N_F, so a missing earth floats to mid–supply.
Peak detector BAT54WS + 1u // 1M (tau 1 s); thresholds 1.02 V (bit A) and 0.31 V (bit B) from a 220k/68k/30k ladder,
with about 100 mV of hysteresis from R33/R34 so the bits cannot chatter near a threshold.
Correct: A=1 B=1. Swapped L/N: A=0 B=0. No earth: A=0 B=1. Each arm must withstand a 6 kV impulse: VR37–class HV resistors.



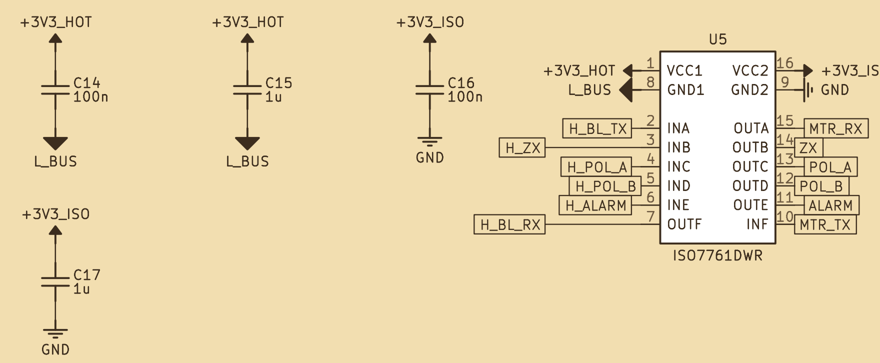
REINFORCED BARRIER, 8 mm

BARRIER CROSSING

The only parts that bridge the barrier.

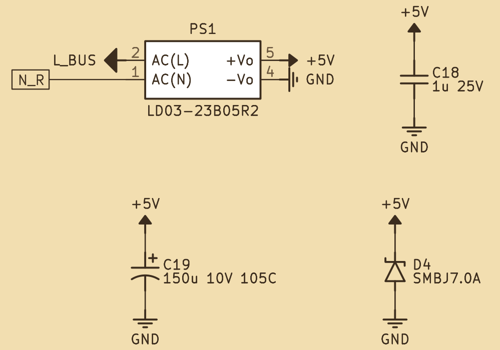
6. DIGITAL ISOLATOR (ISO7761DWR), STRADDLES THE 8 MM BARRIER

Side 1 (hot): BL0939 TX, zero–cross, polarity A/B, alarm in; BL0939 RX out. Side 2 (safe): firmware nets.
Routed slot under the package. Outputs default high if the other side is unpowered: firmware reads POL_x only while the meter is alive.



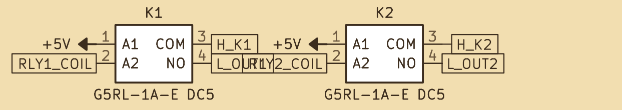
7. SAFE–SIDE 5 V SUPPLY (MORNSUN LD03–23B05R2), STRADDLES THE BARRIER

Genuine Mornsun only. 600 mA, derated to 480 mA at 85 C. Output TVS protects the safe side if the module fails.



8. OUTLET RELAYS, COIL ON THE SAFE SIDE, CONTACTS ON THE HOT SIDE

Live–only switching (ADR–0002). 8 mm between coil and contact pins with a routed slot. Coil 5 V, 80 mA pull–in.

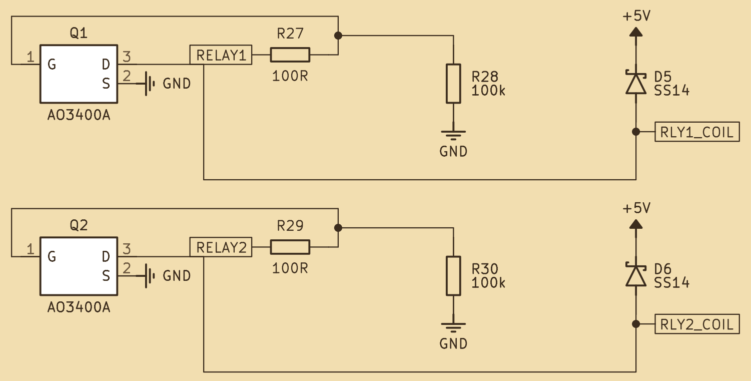


SAFE ZONE

SELV.

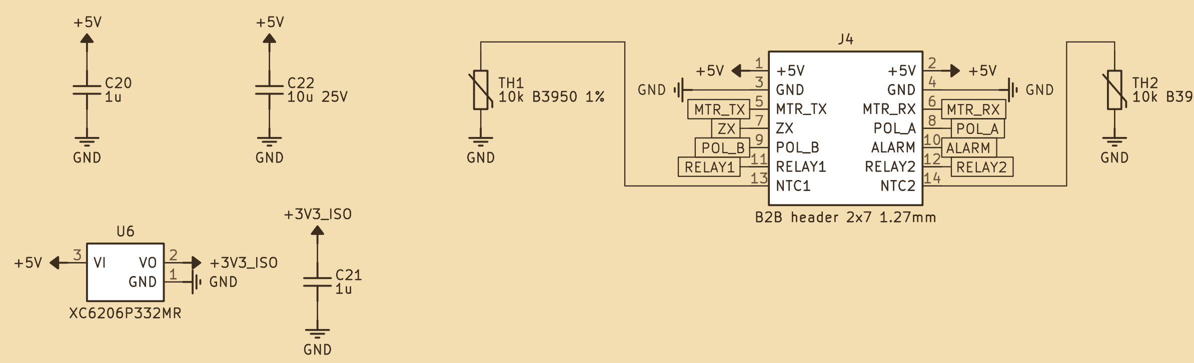
9. RELAY DRIVERS

LEDC PWM at 20 kHz: 100% for 50 ms, then the holding duty. 100k keeps the coil off while the ESP32 boots.



10. ISOLATOR 3.3 V, RELAY NTCS, BOARD–TO–BOARD HEADER

XC6206 feeds only the isolator's safe side (keeps 3.3 V logic levels to the ESP32). NTCs are pressed against each relay body;
their 10k pull–ups sit on the front board next to the ADC.



BOARD NOTES

REAR BOARD, EVT1 DRAFT FOR REVIEW
2 layers, 2 oz. One reinforced barrier: 8.0 mm creepage and clearance, crossed only by U5, PS1, K1, K2.
Net classes: MAINS = L_BUS, N_IN, E_IN, N_F, N_R, L_OUT1/2 and every H_* net. SAFE = everything else.
MAINS to SAFE >= 8.0 mm; MAINS to E_IN >= 3.0 mm; L to N >= 3.0 mm with slots where traces converge.
Outlet paths sized for 20 A combined (IPC–2152). No connector on the hot side.
Bench rule: never power this board from the mains without an isolation transformer and an RCD.

ERC power flags

Hot side is referenced to LIVE (L_BUS). Treat every hot–side net as mains.
Smart Wall Socket. DRAFT for engineering review, not a release.
Aether Labs
Sheet: /
File: sws–rear.kicad_sch
Title: Smart Wall Socket EVT1, rear board
Size: User
Date: 2026–11–10
Rev: EVT1–draft–C
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4
Id: 1/1